GATE
CS & DA

Artificial Intelligence

Proposition & Predictive Logic

Lecture - 01

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Recap of Previous Lecture

Topic — Propositional logic

Topic — Logical Connectives

Topic — Tautology.

Topic — atomic/Compount-proposition

Topic — Turn on Slide map

Topics to be Covered

Topic ⇒ Contradictions

Topic ⇒ Contingency

Topic ⇒ Arguments / Inference

Topic ⇒ Rule of Inference.

Topic

Today is your opportunity to build the tomorrow you want.
—KEN POIROT
RD

Tautology

- A tautology is a compound proposition that is always true, regardless of the truth values of its atomic propositions. In other words, no matter what truth values are assigned to the individual propositions, the overall expression will evaluate to true.

Tautology

- Example $P \vee (P \wedge Q) \leftrightarrow P$

Tautology

- Example $\neg(P \wedge Q) \leftrightarrow (\neg P \vee \neg Q)$

Tautology

- Example $P \rightarrow (Q \vee \neg Q)$

Contradiction

- Contradiction in Propositional Logic
- A contradiction is a compound proposition that is always false, regardless of the truth values of its atomic propositions. In other words, no matter what truth values are assigned to the individual propositions, the overall expression will evaluate to false.

Contradiction

- A contradiction is the opposite of a tautology. It reflects a logical impossibility—something that can never be true. Contradictions are useful in logical reasoning to identify when an argument or a set of premises is inherently flawed or inconsistent.

$(P \wedge (P \rightarrow Q) \wedge \neg Q)$

$P \wedge (P \rightarrow Q) \wedge \neg Q$ (Contraction)

P	Q	$P \rightarrow Q$	$\neg Q$	$P \wedge (P \rightarrow Q) \wedge \neg Q$
F	F	T	T	F
F	T	T	F	F
T	F	F	T	F
T	T	T	F	F

$(\neg(P \rightarrow Q) \wedge P \wedge Q)$

Contradiction.

P	Q	$P \rightarrow Q$	$\neg(P \rightarrow Q)$	$P \wedge Q \wedge \neg(P \rightarrow Q)$
F	F	T	F.	F
F	T	T	F.	F
T	F.	F	T	F
T	T	T	F.	F

What is (P→ Q)∧ (Q→P)

P	Q	P→Q	Q→P	(P→Q)∧(Q→P)
F	F	T	T	T
F	T	T	F	F
T	F	F	T	F
T	T	T	T	T

→ P↔Q
equivalance
Bi Conditional

Compound Proposition

always T ⇒ tautology

always F ⇒ Contradiction

T and F both ⇒ Contingency.

- Satisfiable : a Compound proposition is Satisfiable if in truth table we get at least one True value.

(False ← Q1. If a proposition is Satisfiable then it is always Contingent.
when all T)

True ← Q2. " " " " Contingent " " " " Satisfiable.

True ← Q3. " " " " Tautology then " " " ".

True. ← Q4. " " " " Contradiction" " " never Satisfiable

Contingency in Propositional Logic

- A contingency is a compound proposition that is neither always true (a tautology) nor always false (a contradiction). Its truth value depends on the truth values of its atomic propositions, meaning it can be either true or false based on different assignments of truth values.
- Explanation: Contingent propositions are the most common type of propositions in logic because their truth depends on the specific conditions. Unlike tautologies and contradictions, which have fixed truth values, contingencies vary based on the situation.

Contingency in Propositional Logic

- Identifying a Contingency Truth Table: A proposition is a contingency if its truth table contains both true and false outcomes. This means the truth value depends on specific cases. Logical Status: Contingencies are neither universal truths (like tautologies) nor universal falsehoods (like contradictions).

→ Simple: Create a truth table and Check.

Contingency in Propositional Logic

- Example 1: (P ∧ Q)

P	Q	P∧Q
F	F	F
F	T	F
T	F	F
T	T	T

Contingency

If a Compound proposition is neither Tautology nor Contradiction then it is Contingency.

Tautology	Contradiction	Contingency
Always True	Always False	Both T/F
Satisfiable	Not Satisfiable	Satisfiable.

Satisfiable

- In logic, a statement or a set of statements is satisfiable if there exists at least one interpretation (or assignment of truth values to variables) under which all the statements are true.
- Function which is True for at lease one case.
- For example, consider the logical formula: $(A \vee B)$
- This formula is satisfiable because there exists at least one assignment that makes the formula true.

⇒ So any Compound proposition with at least one T ⇒ Satisfiable

Satisfiable

- In contrast, an unsatisfiable formula is one that cannot be true under any interpretation.
- For instance:$(A \wedge \neg A)$ is unsatisfiable.

$(A \wedge \neg A) \Rightarrow$ Contradiction

A	$\neg A$	$A \wedge \neg A$
T	F	F
F	T	F

Logical Implication

- Let P and Q are any two propositional functions.
- Whenever P is true if Q is also true then P logically implies Q.
- P logically implies Q if and only if P→ Q is a tautology.

- If P and Q are two proposition
  - ↳ If $P \rightarrow Q$, is Valid (True)

Logical Implication

- How to solve the problems :

Which of following is Correct ⇒

1) P logically implies (P∨Q)

we have to check that this proposition is T/F

Yes.

Tave

$P \rightarrow (P \vee Q)$

P	Q	P∨Q	P→(P∨Q)
F	F	F	T
F	T	T	T
T	F	T	T
T	T	T	T

Logical Implication

- How to solve the problems :

Which of following is Correct ⇒

1) P logically implies $(P \wedge Q)$

we have to check that this proposition is T/F

→ Not Correct

So Can I say that P always implies $(P \wedge Q)$ ⇒ No

$P \rightarrow (P \wedge Q)$

P	Q	$(P \wedge Q)$	$P \rightarrow (P \wedge Q)$
F	F	F	T
F	T	F	T
T	F	F	F
T	T	T	T

- any propoqition iq true when it iq true in all caqeq

OR

⇒ Propoqition shd be Tautology to become true

Logical Implication

- How to solve the problems :

Which of following is Correct ⇒

$Q(P \vee Q)$ Logically implies $(P \to Q)$ ⇒ False

P	Q	$P \vee Q$	$P \to Q$	$(P \vee Q) \to (P \to Q)$
F	F	F	T	T
F	T	T	T	T
T	F	T	F	F
T	T	T	T	T

Logical Equivalence

- If P and Q are any two propositional functions, then P equivalent to Q is written as $P \equiv Q$
- P and Q are said to be equivalent if and only if they have same truth table.
- $P \equiv Q$ if and only if $P \leftrightarrow Q$ is a tautology.

Lets check these equivalences

Q. $(\sim P \wedge \sim q) \equiv \sim(P \vee q)$. (True/False)

P	Q	$P \vee Q$	$\sim(P \vee Q)$	$\sim P$	$\sim Q$	$\sim P \wedge \sim Q$	$\sim(P \vee Q) \equiv \sim P \wedge \sim Q$
F	F	F	T	T	T	T	T
F	T	T	F	T	F	F	T
T	F	T	F	F	T	F	T
T	T	T	F	F	F	F	T

So $P \to Q$, $P \equiv Q$ } we need to test and if this is always True then only we say P logically implies Q

P logically equivalent to Q.

Lets check these equivalences

1) $A \vee B \equiv B \vee A$
2) $B \wedge A \equiv A \wedge B$

Commutative

3) $A \rightarrow B \equiv B \rightarrow A \Rightarrow$ false

4) $\neg(\neg A) \equiv A$ ✓

5) $A \vee (B \vee C) \equiv (A \vee B) \vee C$
6) $A \wedge (B \wedge C) \equiv (A \wedge B) \wedge C$

associative

7) $A \wedge (B \vee C) \equiv (A \wedge B) \vee (A \wedge C)$

8) $A \vee (B \wedge C) = (A \vee B) \wedge (A \vee C)$

distributive law.

9) $\neg (A \wedge B) \equiv (\neg A \vee \neg B)$

10) $\neg (A \vee B) = (\neg A \wedge \neg B)$

Demorgan.

Q) $(P \wedge T) \equiv P$ True

P	T	$P \wedge T$	$(P \wedge T) \equiv P$
F.	T	F.	T
T.	T	T.	T

Q) $(P \vee T) \equiv P$ False

P	T	$P \vee T$	$(P \vee T) \equiv P$
F.	T	T.	F
T.	T	T.	T

Argument

- The statement that A set of premises (P1,P2,…...,Pn) yields another propostion Q is called an Argument.
- Q is called the conclusion of the Argument.
- So this is trying to say that when P1,P2,P3 ….. Pn are true then Q is also true….

(Simply a set of Statements i.e Premises & Conclusion)

argument ⇒

P_1 :
P_2 :
P_N :

Conclusion:

Argument

- Example-1 :

P1:

Every student of Information Technology studies Data Structures.

P2:

Data structures necessarily contain the study of arguments.

Therefore, every student of Information Technology studies arguments.

Conclusion:

Argument

- Example-2 :

P1.

- Every parent is a mature person.

P2.

- Children should listen to mature people.
- Therefore, every child should listen to their parents.

Concl.

Argument

- Example-3 :

P1 → Every mother is a woman.

P2 → All women are caring.

- Therefore, every mother is caring.

Conc:

So we learn the set up of argument

Argument

- Types of Arguments
- 1. Valid and Invalid Arguments
- Valid Argument: An argument is valid if the conclusion logically follows from the premises. In other words, if all the premises are true, the conclusion must be true as well.
- Validity is about the logical structure of the argument, not the actual truth of the premises or the conclusion.

⇒ Valid Kab hoga ⇒ P:T, Q:T then Conclusion must be true

- P: All boys have wings
- Q: Animals with wings can fly

Conclusion ⇒ If boys have wings then they can fly.

$\Rightarrow$ P: True.

$\Rightarrow$ Q: True.

$\Rightarrow$ Conclusion $P \to Q$ True

$\Rightarrow$ Valid.

Argument

- Premise 1: All humans are mortal.
  Premise 2: Socrates is a human.
  Conclusion: Socrates is mortal.

$\Rightarrow$ (Valid)

Argument

- Invalid Argument: An argument is invalid if the conclusion does not logically follow from the premises, even if the premises are true. An invalid argument has a flaw in its logical structure, meaning the conclusion could be false even if the premises are true.

Process to check validity of argument

1) all Premises are T
2) then check status of Conclusion,
   If Conclusion is always T ⇒ Valid
   else invalid.

Argument

- Premise 1: If it rains, the ground is wet. $\Rightarrow P \to Q$
- Premise 2: The ground is wet. $\Rightarrow Q$

Conclusion: It is raining. $\Rightarrow P$

Invalid Argument.

To check validity we $P_1 : T$ $\quad P \to Q \; T$

$P_2 : T$ $\quad Q \to T$

So Conclusion Can be T/F

So Kya P hamesha T hoga

P	Q	$P \to Q$
F	F	T
F	T	T
T	F	F
T	T	T

Argument

- Methods of checking whether an argument is valid or not
- 1. Make truth table and make all columns of the different premises p1,p2,…….pn and also make a column of Conclusion
- 2. Mark the rows in which all of p1,p2,……..pn (i.e all premises) are true. such rows are called critical rows.
- 3. Then, in the critical rows, check the value of conclusion , if Q is true in all the critical rows, then the conclusion is valid
- Note: 1. There can be more than one critical row.
- 2. If there is anyone critical row in which Q is false, then the argument is invalid.
- 3. For the argument to be valid, ‘Q’ should be confirmed in all the critical rows in the truth table.

Conclusion

Argument

- Method 2: (Tautology Method)
- Just make a truth table in which make the columns of p1,p2,p3.....pn, Q , p1∧p2∧p3.....pn and finally $\boxed{p1 \wedge p2 \wedge p3 \wedge \ldots\ldots pn \rightarrow Q.}$
- if p1∧p2∧.....pn -> Q is a Tautology, then the argument is valid.

$(P_1 \wedge P_2 \wedge P_3 - - - \wedge P_n \rightarrow Q)$ If this is True then Valid argument.

Rule of Inference

- A rule of inference is a logical principle that defines how new statements (conclusions) can be derived from existing statements (premises) in a valid way. These rules are the foundation of logical reasoning, ensuring that conclusions follow logically from premises.

Inference ⇒ Process to find that given argument is valid/invalid.

Rule of Inference

- Modus Ponens (MP):
- Also known as affirming the antecedent, this rule allows you to infer the consequent if the antecedent is true.

- Form: $P \Longrightarrow Q$
- P (Premise P is true)
- Conclusion: Q (Therefore, Q is true)

$P_1 : A \rightarrow B$
$P_2 : A$
$Q : B$

- If $A \rightarrow B$ and A is always True then B is also always True

Rule of Inference

- Example:

  A B

  Premise 1: If it rains, the streets will be wet.
- Premise 2: It is raining. A
- Conclusion: The streets will be wet. B

- Valid argument

Conclusion

A	B	A→B
F	F	T
F	T	T
T	F	F
T	T	T

Rule of Inference

- Modus Tollens (MT):
- Also known as denying the consequent, this rule states that if the consequent is false, the antecedent must also be false.
- Form: $P \Longrightarrow Q$ (If P, then Q)
- $\neg Q$ (Not Q)
- Conclusion: ¬P (Not P)

$P_1 : A \rightarrow B$
$P_2 : \neg B$
Concl: $\neg A$

(If $A \rightarrow B$, and $\neg B$ is true then $\neg A$ is also true)

Rule of Inference

- Example:
- Premise 1: If it rains, the streets will be wet. ($A \rightarrow B$)
- Premise 2: The streets are not wet. ($\neg B$)
- Conclusion: It is not raining. ($\neg A$)

$\Rightarrow$ Valid argument.

Conclusion always T

A	B	$\neg A$	$\neg B$	$A \rightarrow B$
F	F	T	T	T
F	T	T	F	T
T	F	F	T	F
T	T	F	F	T

← Critical Row

Rule of Inference

- Hypothetical Syllogism (HS):
- This rule allows you to infer a direct relationship between two statements if one statement leads to another and that second statement leads to a third.
- Form: $P \Longrightarrow Q$
- $Q \Longrightarrow R$
- Conclusion: $P \Longrightarrow R$

If $A \to B$, $B \to C$ then $A \to C$

Rule of Inference

- Example:
- Premise 1: If I study, I will pass the exam.
- Premise 2: If I pass the exam, I will graduate.
- Conclusion: If I study, I will graduate.

A B C

B C

A C

P_1: $A \to B$

P_2: $B \to C$

Conc.: $A \to C$

A	B	C
F	F	F
F	F	T
F	T	F
F	T	T
T	F	F

A	B	C	P_1 $A \to B$	P_2 $B \to C$	Conc. $A \to C$
F	F	F	T	T	T
F	F	T	T	T	T
F	T	F	T	F	T
F	T	T	T	T	T
T	F	F	F	T	F
T	F	T	F	T	F
T	T	F	T	F	T
T	T	T	T	T	T

Rule of Inference

- 4. Disjunctive Syllogism (DS):
- This rule is used when one of two possibilities is true, and one of them is false, allowing you to infer the other must be true.
- Form: $P \vee Q$ (Either P or Q)
- $\neg P$ (Not P)
- Conclusion: Q (Therefore, Q)

Rule of Inference

- Example:
- Premise 1: Either I will have coffee or tea.
- Premise 2: I will not have coffee.
- Conclusion: I will have tea.

Rule of Inference

- Addition:
- This rule allows you to add a disjunction (logical OR) with any statement.
- Form: P
- Conclusion: $P \vee Q$

Rule of Inference

- Example:
- Premise: The sun is shining.
- Conclusion: The sun is shining, or it is raining.

Rule of Inference

- Simplification: This rule allows you to infer one part of a conjunction.
- Form: $P \wedge Q$ (Both P and Q are true)
- Conclusion: P

Rule of Inference

- Example:
- Premise: I have a pen and a notebook.
- Conclusion: I have a pen.

Rule of Inference

- . Conjunction:
- This rule allows you to combine two true statements into a single conjunction.
- Form:P
- Q
- Conclusion: $P \wedge Q$

Rule of Inference

- Example:
- Premise 1: I have a pen.
- Premise 2: I have a notebook.
- Conclusion: I have a pen and a notebook.

Rule of Inference

- Resolution:
- A rule used in disjunctive reasoning to infer new conclusions from disjunctions.
- Form: $P \vee Q$
- $\neg P \vee R$
- Conclusion: $Q \vee R$

Rule of Inference

- Example:
- Premise 1: I will go for a walk or stay home.
- Premise 2: I will not go for a walk or I will read a book.
- Conclusion: I will stay home or read a book.

Rule of Inference

- Constructive Dilemma:
- This rule applies when two conditionals are true, and their disjunction implies a disjunction of their conclusions.
- Form:
- $(P \Longrightarrow Q) \wedge (R \Longrightarrow S)$
- $P \vee R$
- Conclusion: $Q \vee S$

Rule of Inference

- Example:
- Premise 1: If I study, I will pass.
- Premise 2: If I sleep well, I will feel rested.
- Premise 3: Either I will study or sleep well.
- Conclusion: I will either pass or feel rested.

THANK - YOU